Satellite Image Change Detection by pixel analysis using Digital Image Processing Techniques

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Abstract :

Image change detection is an important application of digital image processing that involves identifying variations between two images of the same scene acquired at different points in time. This project presents a pixel-based approach for detecting such changes. The method consists of grayscale conversion, absolute difference computation, binary thresholding, and morphological opening. The resulting binary mask is used to identify changed pixels, while an overlay image provides a visual representation of the detected regions. The method is implemented in Python using OpenCV and NumPy and is demonstrated using two satellite-image datasets representing different urban areas. The effect of threshold selection is also examined to understand its influence on the proportion of pixels classified as changed. The project has two example experiment for a constant area region image one in Bengaluru India and other in NYC with different area under coverage and different time intervals. In the Bengaluru experiment, a threshold value of 30 resulted in 6.07% of pixels being classified as changed, while the corresponding New York City experiment produced a detected pixel-change percentage of 6.22%. The results demonstrate the practical implementation of a simple and computationally accessible image change-detection pipeline. However, pixel-level differences may also result from variations in illumination, shadows, atmospheric conditions, image alignment, and surface appearance. Therefore, the detected regions are interpreted as candidate changes rather than confirmed physical changes. The project provides a practical study of fundamental image-processing operations and their combined application to temporal image analysis.

Keywords: Digital image processing, image change detection, pixel differencing, grayscale conversion, thresholding, morphological opening, satellite imagery, Python.

1. Introduction :

Digital image processing involves the application of computational techniques to manipulate, analyze, and extract meaningful information from digital images. It is widely used in applications such as image enhancement, segmentation, object detection, medical imaging, remote sensing, and computer vision. One relevant application is image change detection, which identify differences between images captured at different points in time.

Image change detection can be performed using different approaches depending on the nature of the images, the type of changes being investigated, and the computational resources available. Pixel-based methods provide a straightforward approach by comparing corresponding pixel values in two images. Differences in intensity can then be used to identify regions that may have changed in between the timeline.

If corresponding pixels in two comparable images should have similar intensity values when their appearance remains unchanged. Significant differences between these values may indicate a change in the scene. Consequently, an effective implementation requires suitable preprocessing and careful selection of processing parameters.

In this project, a classical image-processing pipeline is developed to detect changes between two images acquired at different times. The input images are first converted from color to grayscale. An absolute difference operation is then applied to obtain a difference image representing pixel-wise intensity variations. A threshold is used to convert this difference image into a binary mask, separating pixels with relatively large differences from those with smaller differences. Morphological opening is subsequently applied to reduce small isolated detections. Finally, the resulting mask is used to generate a visual overlay highlighting the detected regions.

The proposed method is implemented using Python, OpenCV, and NumPy. Its operation is demonstrated through two independent satellite-image experiments involving Bengaluru and New York City. The project investigates the effect of changing the threshold value on the proportion of pixels classified as changed.

The main objectives of this project are:

1. To understand the fundamental principles of image change detection using digital image processing.

2. To implement grayscale conversion, absolute differencing, thresholding, and morphological operations in a unified pipeline.
3. To generate difference images, binary masks, and overlays for visual analysis.
4. To examine the sensitivity of detected pixel-change percentages to threshold value selection.
5. To identify the limitations of basic pixel-based comparison and discuss possible improvements.

2. Theoretical Background and Related Work [1][2][3]

2.1 Digital Image Representation

A digital image is a two-dimensional representation of a visual scene composed of discrete picture elements known as pixels. Each pixel stores numerical information describing the appearance of a small region of the image. In a grayscale image, this information is typically represented by a single intensity value, whereas a color image commonly contains three channels representing red, green, and blue (RGB). For an 8-bit grayscale image, the intensity of a pixel generally ranges from 0 to 255, where 0 represents black and 255 represents white. Intermediate values represent different shades of gray. In an RGB image, each channel independently stores an intensity value within this range.

2.2 Image Change Detection

Image change detection is the process of identifying differences between two or more images representing the same scene at different points in time. The objective is to determine which pixels or regions exhibit variations that may indicate changes in the observed scene.Change-detection methods can be broadly categorized into pixel-based, object-based, classification-based, statistical, and learning-based approaches. The present project uses a pixel-based approach because it provides a straightforward way to study temporal image differences using fundamental digital image-processing operations.

2.3 Grayscale Conversion

RGB images are converted into grayscale to simplify processing by representing each pixel with a single intensity value.

[$I = 0.299R + 0.587G + 0.114B$]

2.4 Absolute Image Differencing

The absolute difference between corresponding pixels is calculated to highlight intensity variations between the two images.

[$D(x,y) = | I_2(x,y) - I_1(x,y) |$]

2.5 Threshold-Based Segmentation

A threshold classifies pixels as changed or unchanged based on their difference intensity.

[$M(x,y) = 255$ when $D(x,y)>T$ & $M(x,y) = 0$ when $D(x,y) <= T$]

Here, (T) represents the selected threshold value.

2.6 Morphological Opening

Morphological opening, using erosion followed by dilation, removes small isolated regions from the binary mask. This project uses a (3\times3) kernel to reduce minor noise.

2.7 Related Work

Previous studies have examined pixel-based, object-based, and deep-learning approaches to image change detection [1–3]. This project uses pixel-based differencing because it is simple to implement, computationally efficient, and suitable for demonstrating fundamental digital image processing techniques.

3. Proposed Methodology

3.1 Overview of the Proposed Method

The proposed method detects changes between two satellite images captured at different times. It uses grayscale conversion, absolute image differencing,

thresholding, and morphological opening to generate a change mask and highlighted output.

3.2 Input Image Preparation

Two RGB images of the same area are provided as input. The images must have matching dimensions and channels for pixel-wise comparison. Accurate spatial alignment is also important for reliable results.

3.3 Image Differencing

Both images are converted to grayscale, and the absolute difference is calculated for each corresponding pixel. Larger difference values indicate stronger intensity variations between the images.

3.4 Thresholding and Noise Removal

A fixed threshold of 30 is applied to create a binary change mask. Pixels exceeding this threshold are marked as changed. Morphological opening with a (3\times3) kernel is then used to remove small isolated regions.

3.5 Change Percentage Calculation

The percentage of detected changes is calculated by dividing the number of white pixels in the cleaned mask by the total number of pixels.

[Nc/N x 100]

Here, (Nc) is the number of detected changed pixels and (N) is the total number of pixels.

3.6 Change Visualization

The detected regions are highlighted in red on the second image. The method produces three outputs: the absolute difference image, the binary change mask, and the highlighted overlay.

4. Implementation and Experimental Setup

4.1 Software and Libraries

The project is implemented in Python using OpenCV for image processing and NumPy for numerical operations

4.2 Experimental Images

The method was evaluated using two satellite-image datasets representing different geographical areas. Each experiment compares images of the same area captured at different times. The images are processed independently using the same detection pipeline. The satellite images for both Bengaluru and New York City (NYC) were obtained from the Copernicus Data Space Browser [4] in TIFF format. For Bengaluru, Sentinel-2 Level-2A true-colour images using bands B04, B03, and B02 were selected for 18 January 2023 and 22 January 2026. For NYC, images from January 2020 and January 2026 were used. The image pairs were processed independently using the same change detection pipeline.

4.3 Experimental Parameters

A fixed threshold of **30** was selected as the baseline for identifying pixel differences. A (3\times3) kernel was used for morphological opening. Additional experiments with thresholds of 15 and 50 were conducted to examine the effect of threshold selection on the detected change percentage.

4.4 Testing and Validation

The implementation was tested using three automated test cases: identical images, images containing known differences, and images with mismatched dimensions. The tests verify zero detected change for identical images, valid outputs for changed images, and appropriate error handling for incompatible inputs.

4.5 Reproducibility

The source code, experimental results, and test cases are available in the project repository:

https://github.com/Shashwat26526/Satellite_Image_Change_Detection

5.1 Experimental Example

An experimental study was conducted using Sentinel-2 Level-2A true-colour satellite images of an approximately ($6km^2$) area near Electronic City Phase 2 East and Veerasandra Industrial Estate, Bengaluru. Images captured on 18 January 2023 and 22 January 2026 were obtained from the Copernicus Data Space Browser. The images used 10 m /pixel spatial resolution and had

dimensions of (290x215) pixels with three RGB channels. Both images were processed using the proposed change detection pipeline to generate a difference image, binary change mask, and highlighted overlay.

5.2 Change Detection Results

The proposed method was tested on satellite images from Bengaluru and New York City (NYC). Using a threshold of 30 and a (3\times3) morphological kernel, the method generated difference images, binary change masks, and red-highlighted overlays.

For Bengaluru, the detected change percentage was **6.07%**, while NYC showed **6.22%**. These values represent pixels classified as changed by the algorithm, not verified real-world changes.

5.3 Threshold Sensitivity Analysis

The effect of threshold selection was evaluated using values of 15, 30, and 50.

Threshold	Bengaluru Change (%)	NYC Change (%)
15	15.63	26.93
30	6.07	6.22
50	2.17	2.42

The results show that lower thresholds detect more pixel differences, while higher thresholds classify fewer pixels as changed. Therefore, threshold selection significantly influences the final output.

5.4 Discussion

The experiments demonstrate that absolute image differencing can identify intensity variations between satellite images and visualize them using binary masks and overlays. However, detected differences may also result from shadows, illumination changes, noise, or imperfect image alignment.

The method provides a simple and computationally efficient baseline for satellite image change detection. Further validation is required before interpreting the detected percentages as confirmed physical changes.

Suggested figures for this section: [4]

- ***Figure 1:*** *Bengaluru — a) abs difference image, b) binary change mask, and c) red highlighted overlay.*

- ***Figure 2:*** *NYC — a) abs difference image, b) binary change mask, c) red highlighted overlay.*

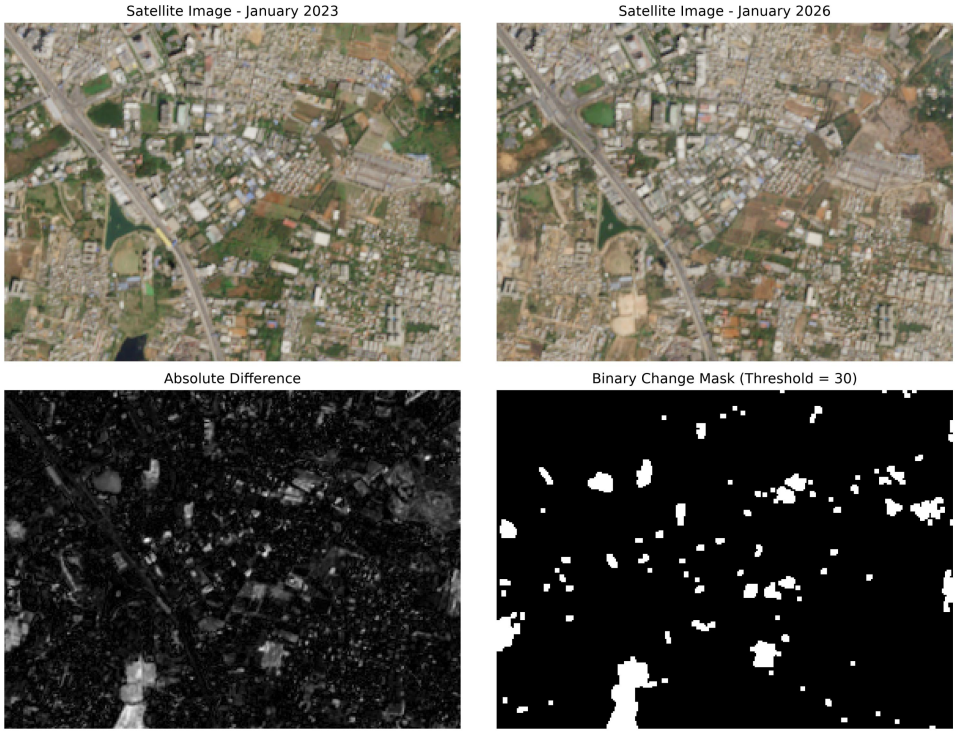


Fig. 1 a),b)

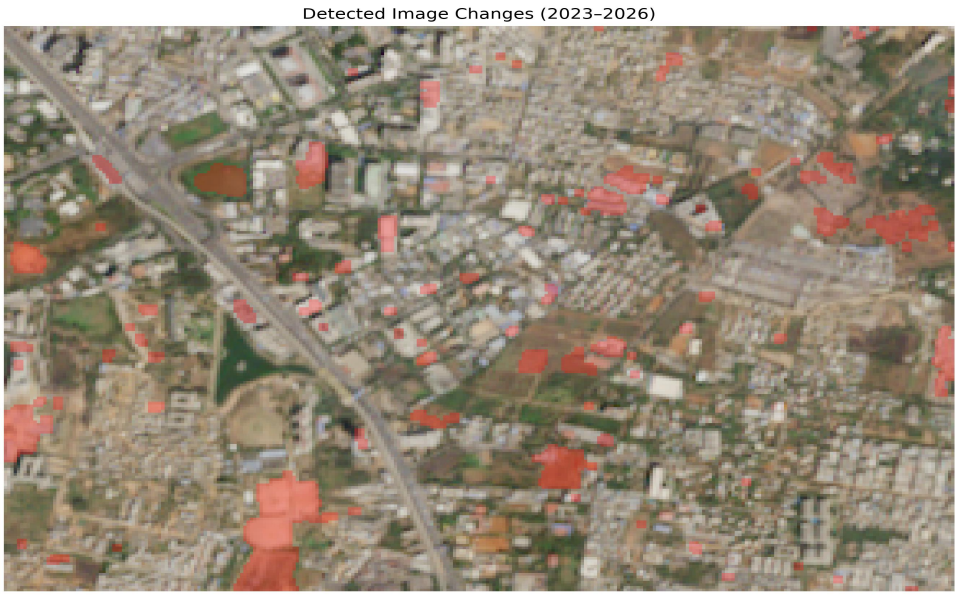


Fig. 1 c)



Fig. 2 a)

Grey Scale difference Image(NYC)

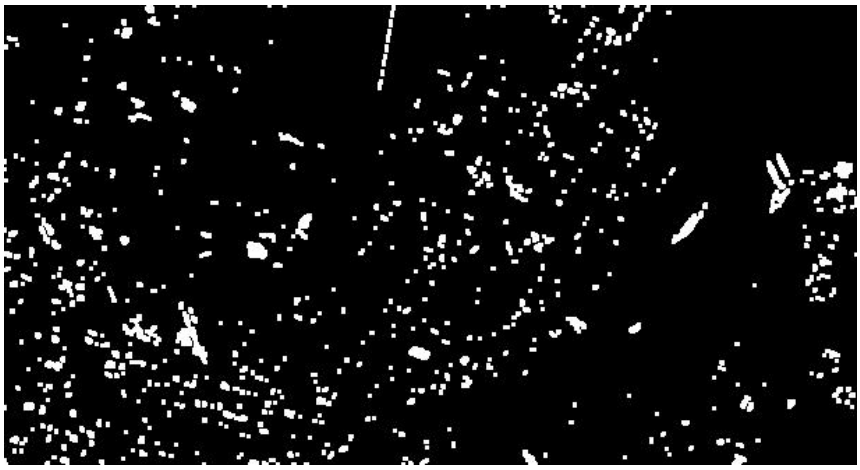


Fig. 2 b)

Binary Masked difference Image (NYC)



Fig 2 c)

Marked Changes Image (NYC)

6. Limitations and Future Improvements

6.1 Limitations

The proposed method is sensitive to threshold selection, illumination differences, shadows, and image misalignment. It detects pixel intensity differences but cannot determine whether they represent meaningful real-world changes. Therefore, the calculated change percentages should be interpreted as detected pixel changes rather than verified physical changes.

6.2 Future Improvements

Future work could incorporate image registration, adaptive thresholding, and advanced change detection techniques to improve reliability. Testing on more datasets with verified reference masks would also help evaluate detection accuracy.

7. Conclusion

This project implemented a satellite image change detection method using grayscale conversion, absolute differencing, thresholding, and morphological opening. Experiments on Bengaluru and New York City images demonstrated that the method can identify and visualize pixel-level differences. Threshold sensitivity analysis showed that the detected change percentage varies considerably with the selected threshold. Although computationally simple, the method is limited by image alignment and environmental variations. Future improvements could make the approach more robust and suitable for broader applications in digital image processing.

8. References

[1] R. J. Radke et al., “Image Change Detection Algorithms: A Systematic Survey,” *IEEE Transactions on Image Processing*, 2005.

[2] M. Hussain et al., “Change Detection from Remotely Sensed Images: From Pixel-Based to Object-Based Approaches,” *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 80, pp. 91–106, 2013.

[3] A. P. Tewkesbury et al., “A Critical Synthesis of Remotely Sensed Optical Image Change Detection Techniques,” *Remote Sensing of Environment*, vol. 160, pp. 1–14, 2015.

[4] Copernicus Data Space Ecosystem, “Copernicus Browser.” Available: https://dataspace.copernicus.eu/browser/ (accessed October 2026).